Name: Hemanth P

Regno: 192524297

DAA LAB

LAB EXPERIMENTS

EXPERIMENT – 6

INSERTION SORT WITH DUPLICATE ELEMENTS

AIM:

To implement the Insertion Sort algorithm in Python and sort an array containing duplicate elements in ascending order.

ALGORITHM:

1. Start the program.
2. Initialize an array containing duplicate elements.
3. Consider the second element as the key.
4. Compare the key with the elements before it.
5. Shift larger elements one position to the right.
6. Insert the key into its correct position.
7. Repeat until all elements are sorted.
8. Display the sorted array.
9. Stop the program.

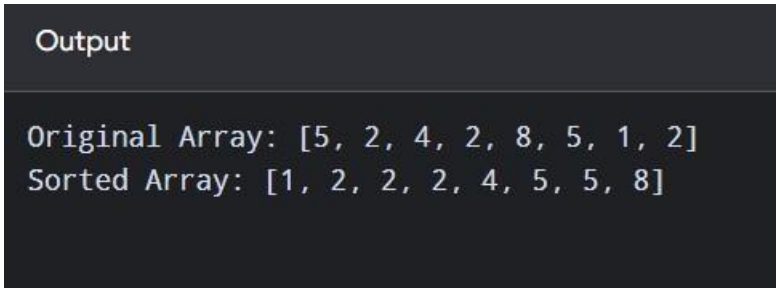
PROCEDURE:

1. Open a Python IDE or editor.
2. Create an array containing duplicate values.
3. Define a function to perform Insertion Sort.
4. Select each element as the key and compare it with previous elements.
5. Shift larger elements and insert the key in the correct position.
6. Execute the program and verify the output.

PYTHON CODE:

```python
def insertion_sort(arr):
    for i in range(1, len(arr)):
        key = arr[i]
        j = i - 1
        while j >= 0 and arr[j] > key:
            arr[j + 1] = arr[j]
            j -= 1
        arr[j + 1] = key
    return arr
arr = [5, 2, 4, 2, 8, 5, 1, 2]
print("Original Array:", arr)
print("Sorted Array:", insertion_sort(arr))
```

SAMPLE OUTPUT:



RESULT:

The Insertion Sort algorithm was successfully implemented in Python. The array containing duplicate elements was sorted in ascending order while preserving all duplicate values.